SG No. 142 D_{4h}^{20} $I4_1/acd$ [tetragonal]

* plus set: $+[0,\ 0,\ 0]$, $+\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$
* Wyckoff site: `8a`, site symmetry: `-4..`

Table 1: Wyckoff bond: `8a@8a`, bond symmetry: `-4..`

No.	vector	center	mapping
1	$[0,\ 0,\ Z]$	$\left[0,\ \frac{1}{4},\ \frac{3}{8}\right]$	`[1,2,-11,-12]`
2	$[0,\ 0,\ Z]$	$\left[0,\ \frac{3}{4},\ \frac{5}{8}\right]$	`[3,4,-9,-10]`
3	$[0,\ 0,\ -Z]$	$\left[\frac{1}{2},\ \frac{1}{4},\ \frac{5}{8}\right]$	`[5,6,-15,-16]`
4	$[0,\ 0,\ -Z]$	$\left[\frac{1}{2},\ \frac{3}{4},\ \frac{3}{8}\right]$	`[7,8,-13,-14]`

Table 2: Wyckoff bond: `16b@8a`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X,\ Y,\ 0]$	$\left[0,\ \frac{1}{4},\ \frac{3}{8}\right]$	`[1,-2]`
2	$[-Y,\ X,\ 0]$	$\left[0,\ \frac{3}{4},\ \frac{5}{8}\right]$	`[3,-4]`
3	$[-X,\ Y,\ 0]$	$\left[\frac{1}{2},\ \frac{1}{4},\ \frac{5}{8}\right]$	`[5,-6]`
4	$[Y,\ X,\ 0]$	$\left[\frac{1}{2},\ \frac{3}{4},\ \frac{3}{8}\right]$	`[7,-8]`
5	$[-X,\ -Y,\ 0]$	$\left[0,\ \frac{3}{4},\ \frac{5}{8}\right]$	`[9,-10]`
6	$[Y,\ -X,\ 0]$	$\left[0,\ \frac{1}{4},\ \frac{3}{8}\right]$	`[11,-12]`
7	$[X,\ -Y,\ 0]$	$\left[\frac{1}{2},\ \frac{3}{4},\ \frac{3}{8}\right]$	`[13,-14]`
8	$[-Y,\ -X,\ 0]$	$\left[\frac{1}{2},\ \frac{1}{4},\ \frac{5}{8}\right]$	`[15,-16]`

Table 3: Wyckoff bond: `32c@8a`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[0,\ \frac{1}{4},\ \frac{3}{8}\right]$	`[1]`
2	$[-X,\ -Y,\ Z]$	$\left[0,\ \frac{1}{4},\ \frac{3}{8}\right]$	`[2]`
3	$[-Y,\ X,\ Z]$	$\left[0,\ \frac{3}{4},\ \frac{5}{8}\right]$	`[3]`
4	$[Y,\ -X,\ Z]$	$\left[0,\ \frac{3}{4},\ \frac{5}{8}\right]$	`[4]`
5	$[-X,\ Y,\ -Z]$	$\left[\frac{1}{2},\ \frac{1}{4},\ \frac{5}{8}\right]$	`[5]`
6	$[X,\ -Y,\ -Z]$	$\left[\frac{1}{2},\ \frac{1}{4},\ \frac{5}{8}\right]$	`[6]`
7	$[Y,\ X,\ -Z]$	$\left[\frac{1}{2},\ \frac{3}{4},\ \frac{3}{8}\right]$	`[7]`
8	$[-Y,\ -X,\ -Z]$	$\left[\frac{1}{2},\ \frac{3}{4},\ \frac{3}{8}\right]$	`[8]`
9	$[-X,\ -Y,\ -Z]$	$\left[0,\ \frac{3}{4},\ \frac{5}{8}\right]$	`[9]`
10	$[X,\ Y,\ -Z]$	$\left[0,\ \frac{3}{4},\ \frac{5}{8}\right]$	`[10]`
11	$[Y,\ -X,\ -Z]$	$\left[0,\ \frac{1}{4},\ \frac{3}{8}\right]$	`[11]`
12	$[-Y,\ X,\ -Z]$	$\left[0,\ \frac{1}{4},\ \frac{3}{8}\right]$	`[12]`
13	$[X,\ -Y,\ Z]$	$\left[\frac{1}{2},\ \frac{3}{4},\ \frac{3}{8}\right]$	`[13]`
14	$[-X,\ Y,\ Z]$	$\left[\frac{1}{2},\ \frac{3}{4},\ \frac{3}{8}\right]$	`[14]`
15	$[-Y,\ -X,\ Z]$	$\left[\frac{1}{2},\ \frac{1}{4},\ \frac{5}{8}\right]$	`[15]`
16	$[Y,\ X,\ Z]$	$\left[\frac{1}{2},\ \frac{1}{4},\ \frac{5}{8}\right]$	`[16]`

* Wyckoff site: `8b`, site symmetry: `2.22`

Table 4: Wyckoff bond: `8a@8b`, bond symmetry: `2.22`

No.	vector	center	mapping
1	$[X, X, 0]$	$[0, \frac{1}{4}, \frac{1}{8}]$	`[1,-2,7,-8]`
2	$[-X, X, 0]$	$[0, \frac{3}{4}, \frac{3}{8}]$	`[3,-4,5,-6]`
3	$[-X, -X, 0]$	$[0, \frac{3}{4}, \frac{7}{8}]$	`[9,-10,15,-16]`
4	$[X, -X, 0]$	$[0, \frac{1}{4}, \frac{5}{8}]$	`[11,-12,13,-14]`

Table 5: Wyckoff bond: `8b@8b`, bond symmetry: `2.22`

No.	vector	center	mapping
1	$[X, -X, 0]$	$[0, \frac{1}{4}, \frac{1}{8}]$	`[1,-2,-7,8]`
2	$[X, X, 0]$	$[0, \frac{3}{4}, \frac{3}{8}]$	`[3,-4,-5,6]`
3	$[-X, X, 0]$	$[0, \frac{3}{4}, \frac{7}{8}]$	`[9,-10,-15,16]`
4	$[-X, -X, 0]$	$[0, \frac{1}{4}, \frac{5}{8}]$	`[11,-12,-13,14]`

Table 6: Wyckoff bond: `8c@8b`, bond symmetry: `2.22`

No.	vector	center	mapping
1	$[0, 0, Z]$	$[0, \frac{1}{4}, \frac{1}{8}]$	`[1,2,-7,-8]`
2	$[0, 0, Z]$	$[0, \frac{3}{4}, \frac{3}{8}]$	`[3,4,-5,-6]`
3	$[0, 0, -Z]$	$[0, \frac{3}{4}, \frac{7}{8}]$	`[9,10,-15,-16]`
4	$[0, 0, -Z]$	$[0, \frac{1}{4}, \frac{5}{8}]$	`[11,12,-13,-14]`

Table 7: Wyckoff bond: `16d@8b`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, X, Z]$	$[0, \frac{1}{4}, \frac{1}{8}]$	`[1,-8]`
2	$[-X, -X, Z]$	$[0, \frac{1}{4}, \frac{1}{8}]$	`[2,-7]`
3	$[-X, X, Z]$	$[0, \frac{3}{4}, \frac{3}{8}]$	`[3,-6]`
4	$[X, -X, Z]$	$[0, \frac{3}{4}, \frac{3}{8}]$	`[4,-5]`
5	$[-X, -X, -Z]$	$[0, \frac{3}{4}, \frac{7}{8}]$	`[9,-16]`
6	$[X, X, -Z]$	$[0, \frac{3}{4}, \frac{7}{8}]$	`[10,-15]`
7	$[X, -X, -Z]$	$[0, \frac{1}{4}, \frac{5}{8}]$	`[11,-14]`
8	$[-X, X, -Z]$	$[0, \frac{1}{4}, \frac{5}{8}]$	`[12,-13]`

Table 8: Wyckoff bond: `16e@8b`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, -X, Z]$	$[0, \frac{1}{4}, \frac{1}{8}]$	`[1,-7]`
2	$[-X, X, Z]$	$[0, \frac{1}{4}, \frac{1}{8}]$	`[2,-8]`
3	$[X, X, Z]$	$[0, \frac{3}{4}, \frac{3}{8}]$	`[3,-5]`
4	$[-X, -X, Z]$	$[0, \frac{3}{4}, \frac{3}{8}]$	`[4,-6]`
5	$[-X, X, -Z]$	$[0, \frac{3}{4}, \frac{7}{8}]$	`[9,-15]`
6	$[X, -X, -Z]$	$[0, \frac{3}{4}, \frac{7}{8}]$	`[10,-16]`

continued ...

Table 8

No.	vector	center	mapping
7	$[-X, -X, -Z]$	$[0, \frac{1}{4}, \frac{5}{8}]$	`[11,-13]`
8	$[X, X, -Z]$	$[0, \frac{1}{4}, \frac{5}{8}]$	`[12,-14]`

Table 9: Wyckoff bond: `16f@8b`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$[0, \frac{1}{4}, \frac{1}{8}]$	`[1,-2]`
2	$[-Y, X, 0]$	$[0, \frac{3}{4}, \frac{3}{8}]$	`[3,-4]`
3	$[-X, Y, 0]$	$[0, \frac{3}{4}, \frac{3}{8}]$	`[5,-6]`
4	$[Y, X, 0]$	$[0, \frac{1}{4}, \frac{1}{8}]$	`[7,-8]`
5	$[-X, -Y, 0]$	$[0, \frac{3}{4}, \frac{7}{8}]$	`[9,-10]`
6	$[Y, -X, 0]$	$[0, \frac{1}{4}, \frac{5}{8}]$	`[11,-12]`
7	$[X, -Y, 0]$	$[0, \frac{1}{4}, \frac{5}{8}]$	`[13,-14]`
8	$[-Y, -X, 0]$	$[0, \frac{3}{4}, \frac{7}{8}]$	`[15,-16]`

Table 10: Wyckoff bond: `32g@8b`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, \frac{1}{4}, \frac{1}{8}]$	`[1]`
2	$[-X, -Y, Z]$	$[0, \frac{1}{4}, \frac{1}{8}]$	`[2]`
3	$[-Y, X, Z]$	$[0, \frac{3}{4}, \frac{3}{8}]$	`[3]`
4	$[Y, -X, Z]$	$[0, \frac{3}{4}, \frac{3}{8}]$	`[4]`
5	$[-X, Y, -Z]$	$[0, \frac{3}{4}, \frac{3}{8}]$	`[5]`
6	$[X, -Y, -Z]$	$[0, \frac{3}{4}, \frac{3}{8}]$	`[6]`
7	$[Y, X, -Z]$	$[0, \frac{1}{4}, \frac{1}{8}]$	`[7]`
8	$[-Y, -X, -Z]$	$[0, \frac{1}{4}, \frac{1}{8}]$	`[8]`
9	$[-X, -Y, -Z]$	$[0, \frac{3}{4}, \frac{7}{8}]$	`[9]`
10	$[X, Y, -Z]$	$[0, \frac{3}{4}, \frac{7}{8}]$	`[10]`
11	$[Y, -X, -Z]$	$[0, \frac{1}{4}, \frac{5}{8}]$	`[11]`
12	$[-Y, X, -Z]$	$[0, \frac{1}{4}, \frac{5}{8}]$	`[12]`
13	$[X, -Y, Z]$	$[0, \frac{1}{4}, \frac{5}{8}]$	`[13]`
14	$[-X, Y, Z]$	$[0, \frac{1}{4}, \frac{5}{8}]$	`[14]`
15	$[-Y, -X, Z]$	$[0, \frac{3}{4}, \frac{7}{8}]$	`[15]`
16	$[Y, X, Z]$	$[0, \frac{3}{4}, \frac{7}{8}]$	`[16]`

* Wyckoff site: `16c`, site symmetry: `-1`

Table 11: Wyckoff bond: `16a@16c`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, 0, 0]$	`[1,-9]`
2	$[-X, -Y, Z]$	$[\frac{1}{2}, 0, \frac{1}{2}]$	`[2,-10]`
3	$[-Y, X, Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[3,-11]`
4	$[Y, -X, Z]$	$[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$	`[4,-12]`

continued ...

Table 11

No.	vector	center	mapping
5	$[-X, Y, -Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[5,-13]`
6	$[X, -Y, -Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[6,-14]`
7	$[Y, X, -Z]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[7,-15]`
8	$[-Y, -X, -Z]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[8,-16]`

* Wyckoff site: `16d`, site symmetry: `2..`

Table 12: Wyckoff bond: `16a@16d`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[0, \frac{1}{4}, z\right]$	`[1,-2]`
2	$[-Y, X, 0]$	$\left[0, \frac{3}{4}, z+\frac{1}{4}\right]$	`[3,-4]`
3	$[-X, Y, 0]$	$\left[\frac{1}{2}, \frac{1}{4}, -z\right]$	`[5,-6]`
4	$[Y, X, 0]$	$\left[\frac{1}{2}, \frac{3}{4}, \frac{3}{4}-z\right]$	`[7,-8]`
5	$[-X, -Y, 0]$	$\left[0, \frac{3}{4}, -z\right]$	`[9,-10]`
6	$[Y, -X, 0]$	$\left[0, \frac{1}{4}, \frac{3}{4}-z\right]$	`[11,-12]`
7	$[X, -Y, 0]$	$\left[\frac{1}{2}, \frac{3}{4}, z\right]$	`[13,-14]`
8	$[-Y, -X, 0]$	$\left[\frac{1}{2}, \frac{1}{4}, z+\frac{1}{4}\right]$	`[15,-16]`

Table 13: Wyckoff bond: `16b@16d`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[0, \frac{1}{4}, z\right]$	`[1,2]`
2	$[0, 0, Z]$	$\left[0, \frac{3}{4}, z+\frac{1}{4}\right]$	`[3,4]`
3	$[0, 0, -Z]$	$\left[\frac{1}{2}, \frac{1}{4}, -z\right]$	`[5,6]`
4	$[0, 0, -Z]$	$\left[\frac{1}{2}, \frac{3}{4}, \frac{3}{4}-z\right]$	`[7,8]`
5	$[0, 0, -Z]$	$\left[0, \frac{3}{4}, -z\right]$	`[9,10]`
6	$[0, 0, -Z]$	$\left[0, \frac{1}{4}, \frac{3}{4}-z\right]$	`[11,12]`
7	$[0, 0, Z]$	$\left[\frac{1}{2}, \frac{3}{4}, z\right]$	`[13,14]`
8	$[0, 0, Z]$	$\left[\frac{1}{2}, \frac{1}{4}, z+\frac{1}{4}\right]$	`[15,16]`

Table 14: Wyckoff bond: `32c@16d`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[0, \frac{1}{4}, z\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[0, \frac{1}{4}, z\right]$	`[2]`
3	$[-Y, X, Z]$	$\left[0, \frac{3}{4}, z+\frac{1}{4}\right]$	`[3]`
4	$[Y, -X, Z]$	$\left[0, \frac{3}{4}, z+\frac{1}{4}\right]$	`[4]`
5	$[-X, Y, -Z]$	$\left[\frac{1}{2}, \frac{1}{4}, -z\right]$	`[5]`
6	$[X, -Y, -Z]$	$\left[\frac{1}{2}, \frac{1}{4}, -z\right]$	`[6]`
7	$[Y, X, -Z]$	$\left[\frac{1}{2}, \frac{3}{4}, \frac{3}{4}-z\right]$	`[7]`
8	$[-Y, -X, -Z]$	$\left[\frac{1}{2}, \frac{3}{4}, \frac{3}{4}-z\right]$	`[8]`
9	$[-X, -Y, -Z]$	$\left[0, \frac{3}{4}, -z\right]$	`[9]`
10	$[X, Y, -Z]$	$\left[0, \frac{3}{4}, -z\right]$	`[10]`

continued ...

Table 14

No.	vector	center	mapping
11	$[Y, -X, -Z]$	$\left[0, \frac{1}{4}, \frac{3}{4} - z\right]$	`[11]`
12	$[-Y, X, -Z]$	$\left[0, \frac{1}{4}, \frac{3}{4} - z\right]$	`[12]`
13	$[X, -Y, Z]$	$\left[\frac{1}{2}, \frac{3}{4}, z\right]$	`[13]`
14	$[-X, Y, Z]$	$\left[\frac{1}{2}, \frac{3}{4}, z\right]$	`[14]`
15	$[-Y, -X, Z]$	$\left[\frac{1}{2}, \frac{1}{4}, z + \frac{1}{4}\right]$	`[15]`
16	$[Y, X, Z]$	$\left[\frac{1}{2}, \frac{1}{4}, z + \frac{1}{4}\right]$	`[16]`

* Wyckoff site: `16e`, site symmetry: `.2.`

Table 15: Wyckoff bond: `16a@16e`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[0, X, Z]$	$\left[x, 0, \frac{1}{4}\right]$	`[1,-6]`
2	$[0, -X, Z]$	$\left[\frac{1}{2} - x, 0, \frac{3}{4}\right]$	`[2,-5]`
3	$[-X, 0, Z]$	$\left[\frac{1}{4}, x + \frac{3}{4}, \frac{1}{2}\right]$	`[3,-7]`
4	$[X, 0, Z]$	$\left[\frac{1}{4}, \frac{1}{4} - x, 0\right]$	`[4,-8]`
5	$[0, -X, -Z]$	$\left[-x, 0, \frac{3}{4}\right]$	`[9,-14]`
6	$[0, X, -Z]$	$\left[x + \frac{1}{2}, 0, \frac{1}{4}\right]$	`[10,-13]`
7	$[X, 0, -Z]$	$\left[\frac{3}{4}, \frac{1}{4} - x, \frac{1}{2}\right]$	`[11,-15]`
8	$[-X, 0, -Z]$	$\left[\frac{3}{4}, x + \frac{3}{4}, 0\right]$	`[12,-16]`

Table 16: Wyckoff bond: `16b@16e`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[x, 0, \frac{1}{4}\right]$	`[1,6]`
2	$[-X, 0, 0]$	$\left[\frac{1}{2} - x, 0, \frac{3}{4}\right]$	`[2,5]`
3	$[0, X, 0]$	$\left[\frac{1}{4}, x + \frac{3}{4}, \frac{1}{2}\right]$	`[3,7]`
4	$[0, -X, 0]$	$\left[\frac{1}{4}, \frac{1}{4} - x, 0\right]$	`[4,8]`
5	$[-X, 0, 0]$	$\left[-x, 0, \frac{3}{4}\right]$	`[9,14]`
6	$[X, 0, 0]$	$\left[x + \frac{1}{2}, 0, \frac{1}{4}\right]$	`[10,13]`
7	$[0, -X, 0]$	$\left[\frac{3}{4}, \frac{1}{4} - x, \frac{1}{2}\right]$	`[11,15]`
8	$[0, X, 0]$	$\left[\frac{3}{4}, x + \frac{3}{4}, 0\right]$	`[12,16]`

Table 17: Wyckoff bond: `32c@16e`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[x, 0, \frac{1}{4}\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{2} - x, 0, \frac{3}{4}\right]$	`[2]`
3	$[-Y, X, Z]$	$\left[\frac{1}{4}, x + \frac{3}{4}, \frac{1}{2}\right]$	`[3]`
4	$[Y, -X, Z]$	$\left[\frac{1}{4}, \frac{1}{4} - x, 0\right]$	`[4]`
5	$[-X, Y, -Z]$	$\left[\frac{1}{2} - x, 0, \frac{3}{4}\right]$	`[5]`
6	$[X, -Y, -Z]$	$\left[x, 0, \frac{1}{4}\right]$	`[6]`
7	$[Y, X, -Z]$	$\left[\frac{1}{4}, x + \frac{3}{4}, \frac{1}{2}\right]$	`[7]`
8	$[-Y, -X, -Z]$	$\left[\frac{1}{4}, \frac{1}{4} - x, 0\right]$	`[8]`

continued ...

Table 17

No.	vector	center	mapping
9	$[-X, -Y, -Z]$	$\left[-x, 0, \frac{3}{4}\right]$	`[9]`
10	$[X, Y, -Z]$	$\left[x+\frac{1}{2}, 0, \frac{1}{4}\right]$	`[10]`
11	$[Y, -X, -Z]$	$\left[\frac{3}{4}, \frac{1}{4}-x, \frac{1}{2}\right]$	`[11]`
12	$[-Y, X, -Z]$	$\left[\frac{3}{4}, x+\frac{3}{4}, 0\right]$	`[12]`
13	$[X, -Y, Z]$	$\left[x+\frac{1}{2}, 0, \frac{1}{4}\right]$	`[13]`
14	$[-X, Y, Z]$	$\left[-x, 0, \frac{3}{4}\right]$	`[14]`
15	$[-Y, -X, Z]$	$\left[\frac{3}{4}, \frac{1}{4}-x, \frac{1}{2}\right]$	`[15]`
16	$[Y, X, Z]$	$\left[\frac{3}{4}, x+\frac{3}{4}, 0\right]$	`[16]`

* Wyckoff site: `16f`, site symmetry: `..2`

Table 18: Wyckoff bond: `16a@16f`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, -X, Z]$	$\left[x, x+\frac{1}{4}, \frac{1}{8}\right]$	`[1,-7]`
2	$[-X, X, Z]$	$\left[\frac{1}{2}-x, \frac{3}{4}-x, \frac{5}{8}\right]$	`[2,-8]`
3	$[X, X, Z]$	$\left[-x, x+\frac{3}{4}, \frac{3}{8}\right]$	`[3,-5]`
4	$[-X, -X, Z]$	$\left[x+\frac{1}{2}, \frac{1}{4}-x, \frac{7}{8}\right]$	`[4,-6]`
5	$[-X, X, -Z]$	$\left[-x, \frac{3}{4}-x, \frac{7}{8}\right]$	`[9,-15]`
6	$[X, -X, -Z]$	$\left[x+\frac{1}{2}, x+\frac{1}{4}, \frac{3}{8}\right]$	`[10,-16]`
7	$[-X, -X, -Z]$	$\left[x, \frac{1}{4}-x, \frac{5}{8}\right]$	`[11,-13]`
8	$[X, X, -Z]$	$\left[\frac{1}{2}-x, x+\frac{3}{4}, \frac{1}{8}\right]$	`[12,-14]`

Table 19: Wyckoff bond: `16b@16f`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, X, 0]$	$\left[x, x+\frac{1}{4}, \frac{1}{8}\right]$	`[1,7]`
2	$[-X, -X, 0]$	$\left[\frac{1}{2}-x, \frac{3}{4}-x, \frac{5}{8}\right]$	`[2,8]`
3	$[-X, X, 0]$	$\left[-x, x+\frac{3}{4}, \frac{3}{8}\right]$	`[3,5]`
4	$[X, -X, 0]$	$\left[x+\frac{1}{2}, \frac{1}{4}-x, \frac{7}{8}\right]$	`[4,6]`
5	$[-X, -X, 0]$	$\left[-x, \frac{3}{4}-x, \frac{7}{8}\right]$	`[9,15]`
6	$[X, X, 0]$	$\left[x+\frac{1}{2}, x+\frac{1}{4}, \frac{3}{8}\right]$	`[10,16]`
7	$[X, -X, 0]$	$\left[x, \frac{1}{4}-x, \frac{5}{8}\right]$	`[11,13]`
8	$[-X, X, 0]$	$\left[\frac{1}{2}-x, x+\frac{3}{4}, \frac{1}{8}\right]$	`[12,14]`

Table 20: Wyckoff bond: `32c@16f`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[x, x+\frac{1}{4}, \frac{1}{8}\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{2}-x, \frac{3}{4}-x, \frac{5}{8}\right]$	`[2]`
3	$[-Y, X, Z]$	$\left[-x, x+\frac{3}{4}, \frac{3}{8}\right]$	`[3]`
4	$[Y, -X, Z]$	$\left[x+\frac{1}{2}, \frac{1}{4}-x, \frac{7}{8}\right]$	`[4]`
5	$[-X, Y, -Z]$	$\left[-x, x+\frac{3}{4}, \frac{3}{8}\right]$	`[5]`
6	$[X, -Y, -Z]$	$\left[x+\frac{1}{2}, \frac{1}{4}-x, \frac{7}{8}\right]$	`[6]`

continued ...

Table 20

No.	vector	center	mapping
7	$[Y,\ X,\ -Z]$	$\left[x,\ x+\frac{1}{4},\ \frac{1}{8}\right]$	[7]
8	$[-Y,\ -X,\ -Z]$	$\left[\frac{1}{2}-x,\ \frac{3}{4}-x,\ \frac{5}{8}\right]$	[8]
9	$[-X,\ -Y,\ -Z]$	$\left[-x,\ \frac{3}{4}-x,\ \frac{7}{8}\right]$	[9]
10	$[X,\ Y,\ -Z]$	$\left[x+\frac{1}{2},\ x+\frac{1}{4},\ \frac{3}{8}\right]$	[10]
11	$[Y,\ -X,\ -Z]$	$\left[x,\ \frac{1}{4}-x,\ \frac{5}{8}\right]$	[11]
12	$[-Y,\ X,\ -Z]$	$\left[\frac{1}{2}-x,\ x+\frac{3}{4},\ \frac{1}{8}\right]$	[12]
13	$[X,\ -Y,\ Z]$	$\left[x,\ \frac{1}{4}-x,\ \frac{5}{8}\right]$	[13]
14	$[-X,\ Y,\ Z]$	$\left[\frac{1}{2}-x,\ x+\frac{3}{4},\ \frac{1}{8}\right]$	[14]
15	$[-Y,\ -X,\ Z]$	$\left[-x,\ \frac{3}{4}-x,\ \frac{7}{8}\right]$	[15]
16	$[Y,\ X,\ Z]$	$\left[x+\frac{1}{2},\ x+\frac{1}{4},\ \frac{3}{8}\right]$	[16]

* Wyckoff site: 32g, site symmetry: 1

Table 21: Wyckoff bond: 32a@32g, bond symmetry: 1

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$[x,\ y,\ z]$	[1]
2	$[-X,\ -Y,\ Z]$	$\left[\frac{1}{2}-x,\ -y,\ z+\frac{1}{2}\right]$	[2]
3	$[-Y,\ X,\ Z]$	$\left[\frac{1}{4}-y,\ x+\frac{3}{4},\ z+\frac{1}{4}\right]$	[3]
4	$[Y,\ -X,\ Z]$	$\left[y+\frac{1}{4},\ \frac{1}{4}-x,\ z+\frac{3}{4}\right]$	[4]
5	$[-X,\ Y,\ -Z]$	$\left[\frac{1}{2}-x,\ y,\ -z\right]$	[5]
6	$[X,\ -Y,\ -Z]$	$\left[x,\ -y,\ \frac{1}{2}-z\right]$	[6]
7	$[Y,\ X,\ -Z]$	$\left[y+\frac{1}{4},\ x+\frac{3}{4},\ \frac{3}{4}-z\right]$	[7]
8	$[-Y,\ -X,\ -Z]$	$\left[\frac{1}{4}-y,\ \frac{1}{4}-x,\ \frac{1}{4}-z\right]$	[8]
9	$[-X,\ -Y,\ -Z]$	$[-x,\ -y,\ -z]$	[9]
10	$[X,\ Y,\ -Z]$	$\left[x+\frac{1}{2},\ y,\ \frac{1}{2}-z\right]$	[10]
11	$[Y,\ -X,\ -Z]$	$\left[y+\frac{3}{4},\ \frac{1}{4}-x,\ \frac{3}{4}-z\right]$	[11]
12	$[-Y,\ X,\ -Z]$	$\left[\frac{3}{4}-y,\ x+\frac{3}{4},\ \frac{1}{4}-z\right]$	[12]
13	$[X,\ -Y,\ Z]$	$\left[x+\frac{1}{2},\ -y,\ z\right]$	[13]
14	$[-X,\ Y,\ Z]$	$\left[-x,\ y,\ z+\frac{1}{2}\right]$	[14]
15	$[-Y,\ -X,\ Z]$	$\left[\frac{3}{4}-y,\ \frac{1}{4}-x,\ z+\frac{1}{4}\right]$	[15]
16	$[Y,\ X,\ Z]$	$\left[y+\frac{3}{4},\ x+\frac{3}{4},\ z+\frac{3}{4}\right]$	[16]